AMTI (NMTC) - 2011

BHASKARA CONTEST - JUNIOR LEVEL

PART - A

1. n is a natural number greater than 1, and $A = \frac{\sqrt{n+1}}{n} + \frac{\sqrt{n+4}}{n+3} + \frac{\sqrt{n+7}}{n+6} + \frac{\sqrt{n+10}}{n+9} + \frac{\sqrt{n+13}}{n+12}$

$$B = \frac{1}{\sqrt{n-1}} + \frac{1}{\sqrt{n+2}} + \frac{1}{\sqrt{n+5}} + \frac{1}{\sqrt{n+8}} + \frac{1}{\sqrt{n+11}}$$

then

(A) A = B (B) A = 2B (C) A < B (D) A > B

2. How many distinct rational numbers (a, b, c, d) are there with $a\log_{10}2 + b\log_{10}3 + c\log_{10}5 + d\log_{10}7 = 2011$.

(A) 0 (B) 1 (C) 5 (D) 2011

3. ABCD is a rectangle in which AB = 8, AD = 9. E is on AD such that DE = 4. H is on BC such that BH = 6. EC and AH cut at G. GF is drawn perpendicular to AD produced. Then GF =

(A) 20 (B) 22 (C) 18 (D) 15

4. The sides of the base of a rectangular parallelepiped are a and b. The diagonal of the parallelepiped is inclined to the base plane at an angle θ. Then the lateral surface area of the solid is

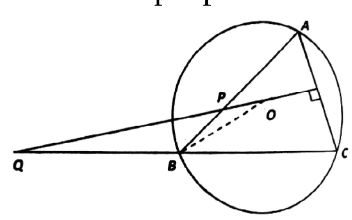
(A) $2(a+b)\sqrt{a^2+b^2}\tan\theta$ (B) $(a+b)\sqrt{a^2+b^2}\tan\theta$

(C) $(a^2+b^2)\sqrt{a+b}\tan\theta$ (D) $2(a^2+b^2)\sqrt{a+b}\tan\theta$

5. The number of integers n which satisfy $(n^2 - 2)(n^2 - 20) < 0$ is

(A) 3 (B) 4 (C) 5 (D) 6

6. In the adjoining figure. O is the circum centre of the triangle ABC. The perpendicular bisector of AC meets AB at P and CB produced at Q. Then



(A) 2∠PQB = 3∠PBO (B) 3∠PQB = 2∠PBO

(C) 4∠PQB = 5∠PBO (D) None of these

7. There are 15 radial spokes in a wheel, all equally inclined to one another. Then there are two spokes which

(A) lie along a diameter of the wheel (B) are perpendicular to each other

(C) are inclined at an angle of 120^0 (D) include an angle less than 24^0

8. Three teams of wood-cutters take part in a competition. The first and the third teams put together produced twice the amount cut by the second team. The second and the third team put together yielded a three-fold output as compared with the first team. Which of the teams won the competition ?

(A) first team (B) second team (C) third team (D) there is a tie

9. The number of positive integral values of n for which $(n^3 - 8n^2 + 20n - 13)$ is a prime number is

(A) 2 (B) 1 (C) 3 (D) 4

10. The value of the expression $\dfrac{\sqrt{(x+2)^2 - 8x}}{(\sqrt{x} - 2/\sqrt{x})}$ is equal to

(A) $\sqrt{x}$ for all x > 0 (B) $-\sqrt{x}$ for 0 < x < 2

(C) $\sqrt{x}$ for 0 < x < 2 (D) $-\sqrt{x}$ for all x > 0

11. The number of positive integers 'n' for which 3n – 4, 4n – 5 and 5n – 3 are all primes is
(A) 1 (B) 2 (C) 3 (D) infinite

12. a and b are the roots of the quadratic equation $x^2 + \lambda x - \frac{1}{2\lambda^2} = 0$ where x is the unknown and λ is a real parameter. The minimum value of $a^4 + b^4$ is
(A) $2\sqrt{2}$ (B) $\frac{1}{1+\sqrt{2}}$ (C) $\sqrt{2}$ (D) $2+\sqrt{2}$

13. When $b \geq 0$, then $12a^2b^3 - a^6 - b^9$
(A) always is less than or equal to 64 (B) always greater than 64
(C) always negative (D) always lies in the interval [60, 64]

14. There are three natural numbers. The second is greater than the first by the amount the third is greater than the second. The product of the two smaller numbers is 85 and the product of the two larger numbers is 115. If the numbers are x, y, z with x < y < z then the value of (2x + y + 8z) is
(A) 117 (B) 119 (C) 121 (D) 78

15. The number of digits in the sum $100 + 100^2 + 100^3 + ... + 100^{2011}$ is
(A) 4023 (B) 4022 (C) 4024 (D) none of these

PART - B

1. In the sequence $a_1, a_2, ..., a_n$ the sum of any three consecutive terms is 40. If the third term is 10 and the eight term in 8 then the 1000th term is ________.
2. ABCDEF is a non-regular hexagon where all the six sides touch a circle and all the six sides are of equal length. If $\angle A = 140^0$ then $\angle D$ = ______.
3. The difference between the largest 6 digit number with no repeated digits and the smallest six digit number with no repeated digits is _______.
4. Three consecutive integers lying between 1000 and 9999, both inclusive, are such that the smallest is divisible by 11 and the middle one by 9 and the largest by 7. The sum of the largest such four digit numbers is _______.
5. f(x) is a quadratic polynomial with f(0) = 6, f(1) = 1 and f(2) = 0. Then f(3) = _____.
6. While multiplying two numbers a and b Renu reverted the digits of a two digit number and obtained the product to be 391. Renu realized that she made a mistake as her correct answer must be even. The correct product is _______.
7. In a chess tournament players get 1 point for a win 0 for a loss and $\frac{1}{2}$ point for a draw. I a tournament where every player plays against every other player exactly once, the top four scores were $5\frac{1}{2}, 4\frac{1}{2}, 4$ and $2\frac{1}{2}$. The lowest score in the tournaments was ______.
8. When x is real, the greatest possible value of $10^2 - 100^x$ is _______.
9. The diagonals of a convex quadrilateral are perpendicular. If AB = 4, AD = 5, CD = 6, then length of BC is _______.
10. Three circles, each of radius one, have centres at A, B and C. Circles A and B touch each other and circle C touches AB at its midpoint. The area inside circle C and outside circles A and B is ______.
11. The number of rectangles that can be obtained by joining four of the 11 vertices of a 11-sided regular polygon is _______.
12. Let D, E, F be the midpoints of the sides BC, CA and AB respectively of triangle ABC. AB = 16, BC = 21 and CA = 19. The circum-circles of the triangles BDF and CDE cut at P other than D. Then $\angle BPC$ = ________.
13. Let x and y be two distinct three digit positive integers such that their average is 600. Then the maximum value of $\frac{x}{y}$ is _______.
14. Let N = 101010...101 by a 2011 digit number with alternating 1's and 0's. The sum of the digits of the product of N with 2011 is _______.
15. x_1, y_1, x_2, y_2 are real numbers. If $x_1^2 + x_2^2 \leq 2$ and $y_1^2 + y_2^2 \leq 4$, the maximum value of the expression $x_1y_1 + x_2y_2$ is _______.
